

Tao LIU

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🌐 <https://ryantaoliu.github.io/>

Research Interests

Neural-based design and manufacturing optimization, computational geometry, additive manufacturing, and robotics. In particular: multi-axis 3D printing, topology optimization, neural implicit representations for geometric modeling, fiber-reinforced composites, and computational design frameworks.

Research Experience

- 2023 – Now **Ph.D. Student**, The University of Manchester, Manchester, UK
Advised by Prof. Charlie C.L. Wang
- 2020 – 2023 **Research Assistant**, Central South University, Changsha, China
Advised by Prof. Shengjun LIU

Collaborative Projects

- Broetje-Automation GmbH & Airbus** – Vector-field guided tool-path planning for 3D printing with CCF (Completed)
- NCC (National Composites Centre)** – Nonplanar Continuous Fibre AM (CFAM) design tool evaluation (Completed)

Research Publications

- [14] Co-optimization of structure and manufacturable semi-continuous layers for laminated composites
Tao LIU, Aoran Lyu, Yongxue Chen, Yu Jiang, Michael Petty, Charlie C.L. Wang
ACM Transactions on Graphics (SIGGRAPH 2026), 45(4), Article 146, 2026. DOI: 10.1145/3811393
- [13] Neural co-optimization of structural topology, manufacturable layers, and path orientations for fiber-reinforced composites
Tao LIU, Tianyu Zhang, Yongxue Chen, Weiming Wang, Yu Jiang, Yuming Huang, Charlie Wang
ACM Transactions on Graphics (TOG), 2025(**Monthly Paper Award, from SoE UoM**)
- [12] Neural Slicer for Multi-Axis 3D Printing
Tao LIU, Tianyu Zhang, Yongxue Chen, Yuming Huang, Charlie Wang
ACM Transactions on Graphics (TOG), 2024
- [11] Curve-Based Slicer for Multi-Axis DLP 3D Printing
Chengkai Dai, **Tao LIU**, Dezhao Guo, Binzhi Sun, Guoxin Fang, Yeung Yam, Charlie C. L. Wang
ACM Transactions on Graphics (TOG), 44(6):1–14, 2025(**Best Paper Award**)
- [10] Can Any Model Be Fabricated? Inverse Operation Based Planning for Hybrid Additive-Subtractive Manufacturing
Yongxue Chen, **Tao LIU**, Yuming Huang, Weiming Wang, Tianyu Zhang, Kun Qian, Zikang Shi, Charlie C. L. Wang
ACM Transactions on Graphics (TOG), 44(6):1–14, 2025
- [9] Toolpath generation for high density spatial fiber printing guided by principal stresses
Tianyu Zhang, **Tao LIU**, Neelotpal Dutta, Yongxue Chen, Renbo Su, Zhizhou Zhang, Weiming Wang, Charlie Wang
Composites Part B: Engineering, 2025

Research Publications (continued)

- [8] Memory-efficient modeling and slicing of large-scale adaptive lattice structures
Shengjun LIU, **Tao LIU**, Qiang Zou, Weiming Wang, Eugeni L Doubrovski, Charlie CL Wang
Journal of Computing and Information Science in Engineering, 21(6):061003, 2021
- [7] Variational progressive-iterative approximation for RBF-based surface reconstruction
Shengjun LIU, **Tao LIU**, Ling Hu, Yuanyuan Shang, Xinru Liu
The Visual Computer, 2021
- [6] TPMS-based mechanical bonding structures optimized by FEM with periodic boundary conditions
Renbo Su, **Tao LIU**, Tianyu Zhang, Yingjun Tian, Romain Hautier, Weiming Wang, Charlie C.L. Wang
Materials & Design, 263:115581, 2026
- [5] Force-Based Adaptive Deposition in Multi-Axis Additive Manufacturing: Low Porosity for Enhanced Strength
Yuming Huang, Renbo Su, Kun Qian, Tianyu Zhang, Yongxue Chen, **Tao LIU**, Guoxin Fang, Weiming Wang, Charlie Wang
Robotics and Computer-Integrated Manufacturing, 2026
- [4] Roadmap on Artificial Intelligence-Augmented Additive Manufacturing
Ali Zolfagharian, Liuchao Jin, Qi Ge, Wei-Hsin Liao, Andrés Díaz Lantada, Francisco Franco Martínez, Tianyu Zhang, **Tao LIU**, Charlie C. L. Wang, and others
Advanced Intelligent Systems, 2026
- [3] Co-Optimization of Tool Orientations, Kinematic Redundancy, and Waypoint Timing for Robot-Assisted Manufacturing
Yongxue Chen, Tianyu Zhang, Yuming Huang, **Tao LIU**, Charlie Wang
IEEE Transactions on Automation Science and Engineering, 2025
- [2] Learning Based Toolpath Planner on Diverse Graphs for 3D Printing
Yuming Huang, Yuhu Guo, Renbo Su, Xingjian Han, Junhao Ding, Tianyu Zhang, **Tao LIU**, Weiming Wang, Guoxin Fang, Xu Song, and others
ACM Transactions on Graphics (TOG), 2024
- [1] Implicit progressive-iterative algorithm of curves and surfaces with compactly supported radial basis functions
Haibo Wang, **Tao LIU**, Shengjun Liu, Wenyan Wei, Xinru Liu, Pingbo Liu, Yanyu Bai, Yue'an Chen
Journal of Computer-Aided Design & Computer Graphics, 2021

Education

2023 – Now	Ph.D. in Neural-Based Design and Manufacturing Optimization , The University of Manchester, Manchester, UK Supervised by Prof. Charlie C.L. Wang
Aug 2014 – Jun 2020	M.S. & B.S. in Computational Mathematics , Central South University, ChangSha, China Supervised by Prof. Shengjun LIU (GPA 3.4/4 top 10%)
Aug 2013 – Jun 2014	B.S. in Applied Chemistry , Central South University, ChangSha, China

Honors & Awards

2025	Best Paper Award , SIGGRAPH Asia Monthly(December) paper award , School of Engineering, University of Manchester <i>Neural co-optimization of structural topology, manufacturable layers, and path orientations for fiber-reinforced composites</i>
2018	Central South University Scholarship , School of Mathematics, Central South University

Honors & Awards (continued)

- 2017 **University Recommendation for M.Sc. Admission (Exam-Exempt)**, School of Mathematics, Central South University
- 2014 **Excellent Student**, School of Chemistry, Central South University

Academic Service

Reviewer for Journals and Conferences

<i>ACM SIGGRAPH and SIGGRAPH Asia</i>	<i>Computer Aided Geometry Design (CAGD)</i>
<i>Additive Manufacturing Process (AMP)</i>	<i>IEEE Transactions on Visualization and Computer Graphics (TVCG)</i>
<i>Computer Aided Design (CAD)</i>	<i>Graphics (TVCG)</i>
<i>ACM Symposium on Computational Fabrication (SCF)</i>	<i>Eurographics (EG)</i>
<i>International Conference on Computer-Aided Design and Computer Graphics (CAD/Graphics)</i>	<i>Graphical Models (GMod)</i>
	<i>Computers & Graphics (CG)</i>

Teaching

SIGGRAPH Asia	Course, Vector3DP: Vector Fields for 3D Printing, https://ryantaoliu.github.io/Vector3DP/
September 2025	Teaching Assistant , Graphics & Virtual Environments, The University of Manchester (Undergraduate course) Teaching Assistant , Fundamentals of Computer Architecture, The University of Manchester (Undergraduate course) Teaching Assistant , Experimental Methods (MSc Structural Engineering), The University of Manchester (Postgraduate course)
September 2024	Teaching Assistant , Introduction to Programming 1, The University of Manchester (Undergraduate course)
September 2023	Teaching Assistant , Additive Manufacturing & 3D Product Modelling, The University of Manchester (Undergraduate course) Teaching Assistant , Advanced Modelling & Simulation, The University of Manchester (Undergraduate course)
September 2018	Teaching Assistant , Computer Graphics, Central South University (Undergraduate course)

References

Available on Request